Loops and trace tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected loop-and-trace record.

Explain the accumulator, predict a range, diagnose a reset and state a domain limit.

Keep one habit: Keep exact range, current value, total before/after, output events and observed status together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Branching and boundary tests

> **My remembered idea, example and correction**

Expressions and operator order

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Warm-up range(1,4): values 1,2,3 and totals 1,3,6; pictured range(1,5): 1,2,3,4 and totals 1,3,6,10. W2 range(2,5): 2,3,4 with totals 2,5,9. Stop excluded. Initialize total 0 before loop; empty range keeps 0. Resetting inside produces only current value in the sum; printing inside records each running total. Default step is positive 1; arbitrary input validation and learning are not implemented.

- List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Warm-up range(1,4): values 1,2,3 and totals 1,3,6; pictured range(1,5): 1,2,3,4 and totals 1,3,6,10. W2 range(2,5): 2,3,4 with totals 2,5,9. Stop excluded. Initialize total 0 before loop; empty range keeps 0. Resetting inside produces only current value in the sum; printing inside records each running total. Default step is positive 1; arbitrary input validation and learning are not implemented.

- Reference for this exercise: open this week's student lesson, "Loops and trace tables", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Warm-up range(1,4): values 1,2,3 and totals 1,3,6; pictured range(1,5): 1,2,3,4 and totals 1,3,6,10. W2 range(2,5): 2,3,4 with totals 2,5,9. Stop excluded. Initialize total 0 before loop; empty range keeps 0. Resetting inside produces only current value in the sum; printing inside records each running total. Default step is positive 1; arbitrary input validation and learning are not implemented.

- Reference for this exercise: open this week's student lesson, "Loops and trace tables", at "Find and repair a mistake". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Warm-up range(1,4): values 1,2,3 and totals 1,3,6; pictured range(1,5): 1,2,3,4 and totals 1,3,6,10. W2 range(2,5): 2,3,4 with totals 2,5,9. Stop excluded. Initialize total 0 before loop; empty range keeps 0. Resetting inside produces only current value in the sum; printing inside records each running total. Default step is positive 1; arbitrary input validation and learning are not implemented.

- In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- List every value visited and each running total. Then predict the result for range(2,2).
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Count iterations and a stopping point

```
total = 0
for value in range(2, 6):
    total = total + value
print(total)
```

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled **loop**s and **trace table**s **evidence** sheet to Python readiness portfolio. Show how you can trace variable values before and after each **loop iteration**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

  Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Accumulate a short range. A running total adds the integers visited by range(1,5).
- Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.
- List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.

AI ACADEMY / PYTHON-BRIDGE/6

Running total after each visited value

	Current value	Running total
Visit 1	1	1
Visit 2	2	3
Visit 3	3	6
Visit 4	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/6

Loops and trace tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Visit the range

Values are 1,2,3,4; the stop value 5 is excluded.

2. Trace running total

Totals after each addition are 1,3,6,10.

3. Print after the loop

The final displayed value is 10.

CHANGE ONE THING

Replace range(1,5) with range(1,1).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project portfolio milestone

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- Accumulate a short range. A running total adds the integers visited by range(1,5).
- Which labelled step matches this detail? The final displayed value is 10.
- 1. Trace running total
- 2. Print after the **loop**
- 3. Visit the range
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Print after the **loop** The final displayed value is 10. A **loop** repeats an operation for each visited value.

A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Changed-case question and hint

- Changed-case practice: Replace range(1,5) with range(1,1). Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. No values are visited and the final **output** is 0.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: No values are visited and the final **output** is 0. The **accumulator** starts at zero and an empty **loop** never changes it.

Repair and keep the record

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

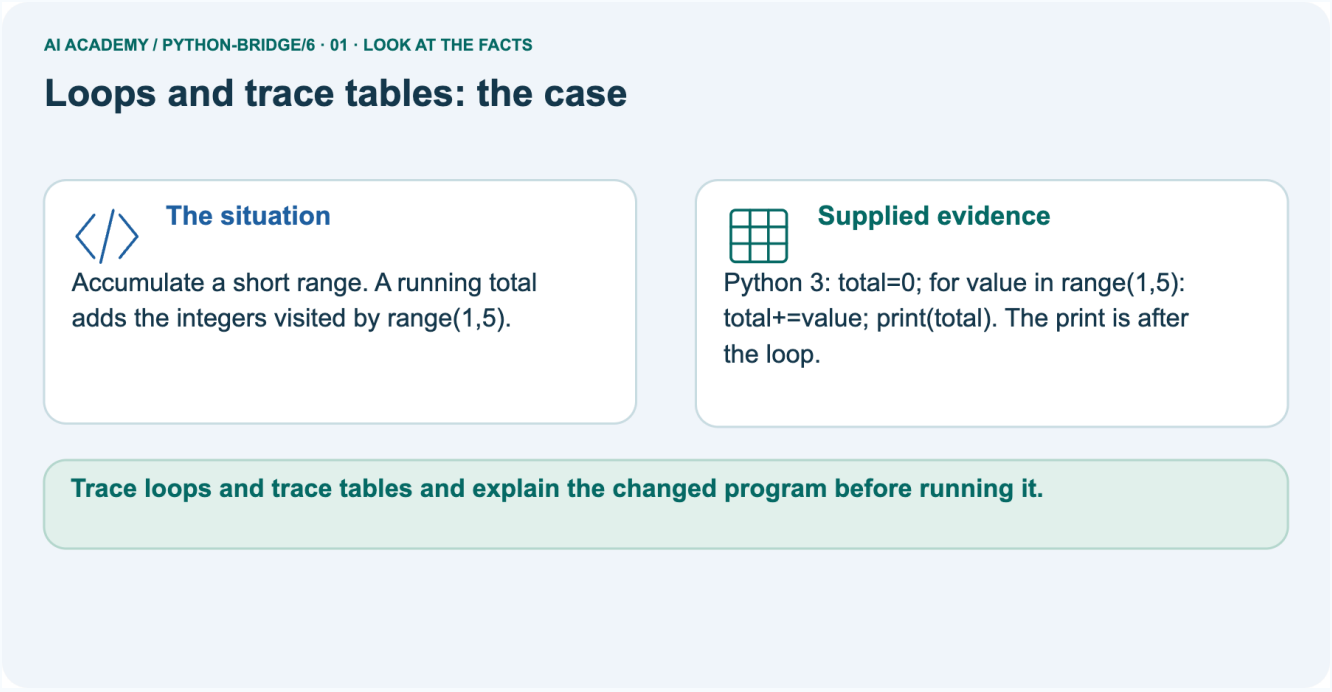
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Loops and trace tables: the case



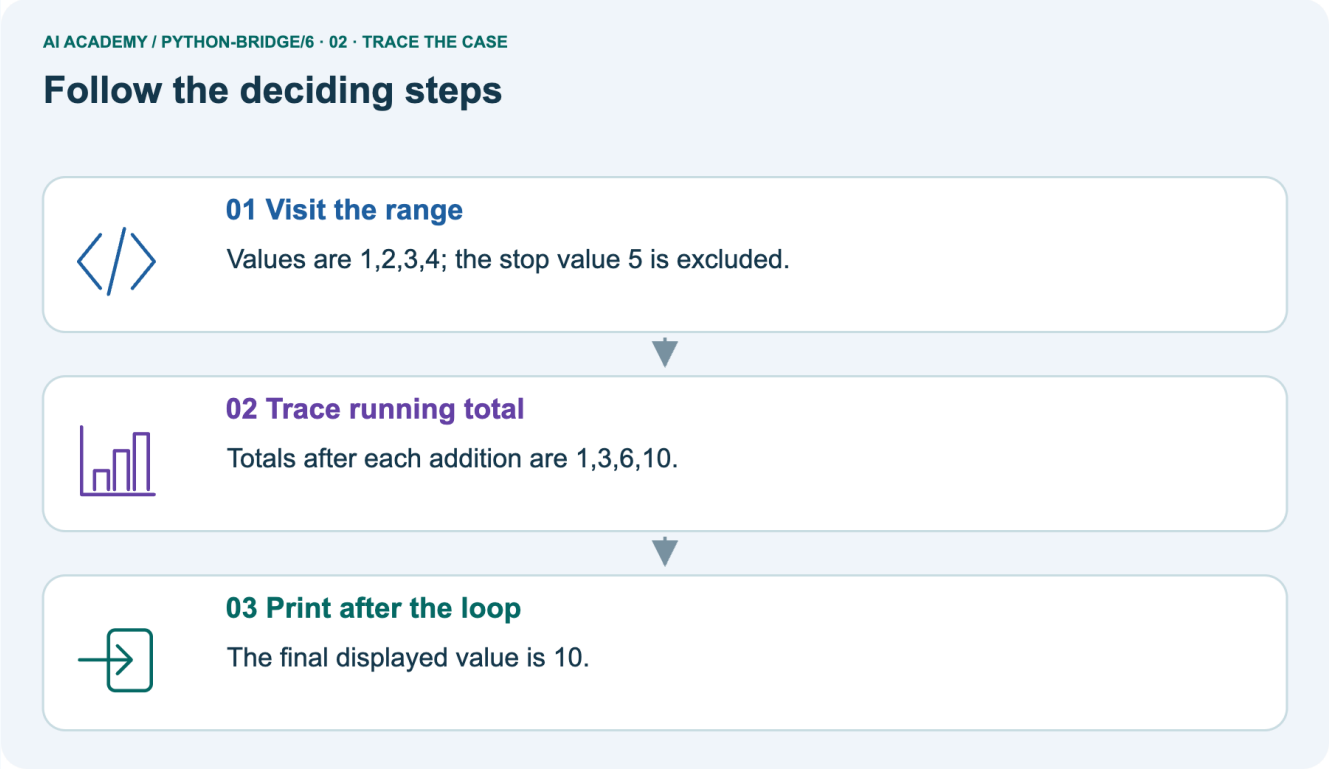
Follow the labelled picture with the matching explanation. Open the vector diagram

Accumulate a short range. A running total adds the integers visited by range(1,5). Supplied facts: Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

Point to the given fact that matters most. Explain why it matters.

> **My picture explanation, deciding evidence and one limit**

02. Follow the deciding steps



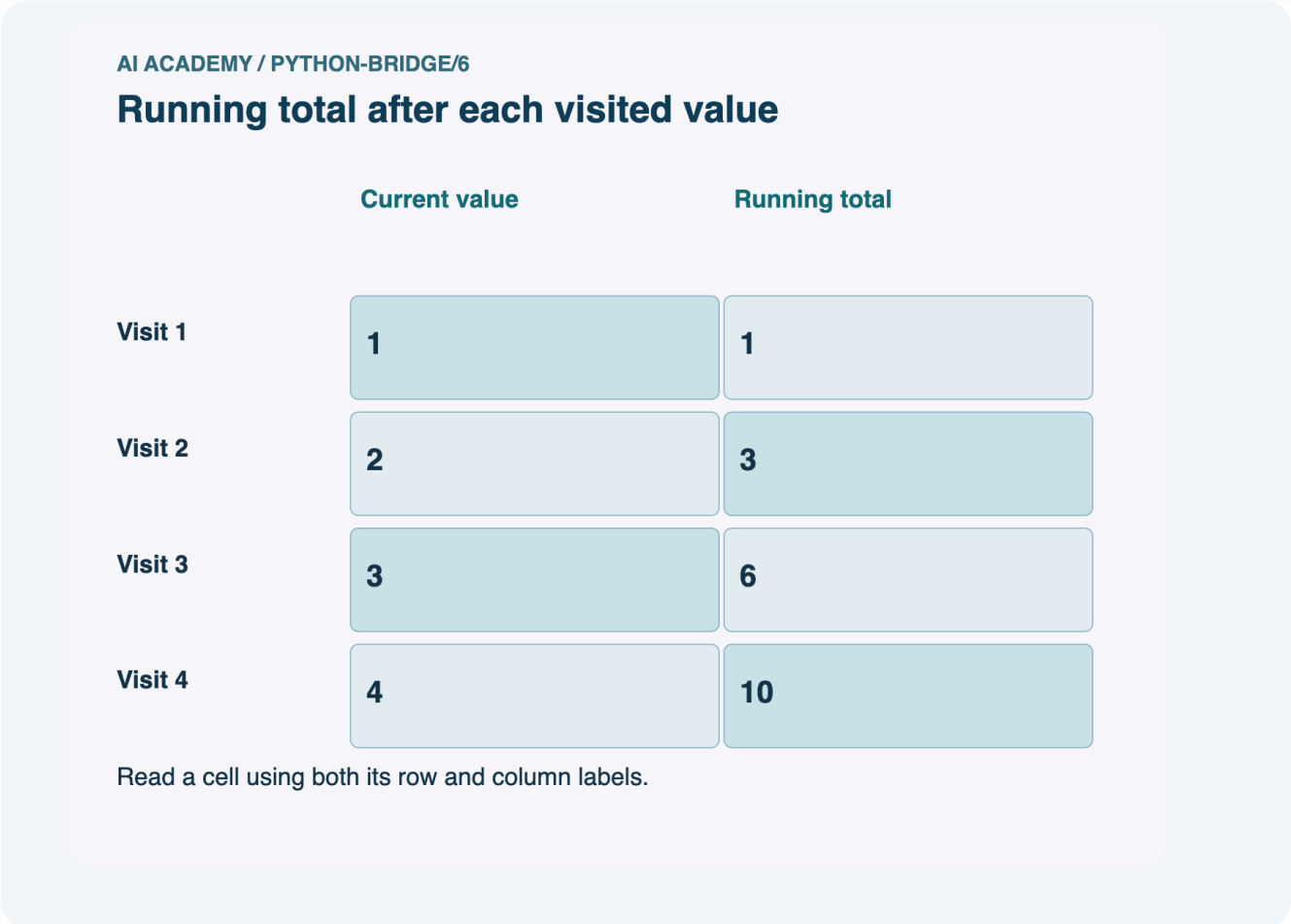
Follow the labelled picture with the matching explanation. Open the vector diagram

Visit the range: Values are 1,2,3,4; the stop value 5 is excluded. → Trace running total: Totals after each addition are 1,3,6,10. → Print after the **loop**: The final displayed value is 10.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Running total after each visited value



Follow the labelled picture with the matching explanation. Open the vector diagram

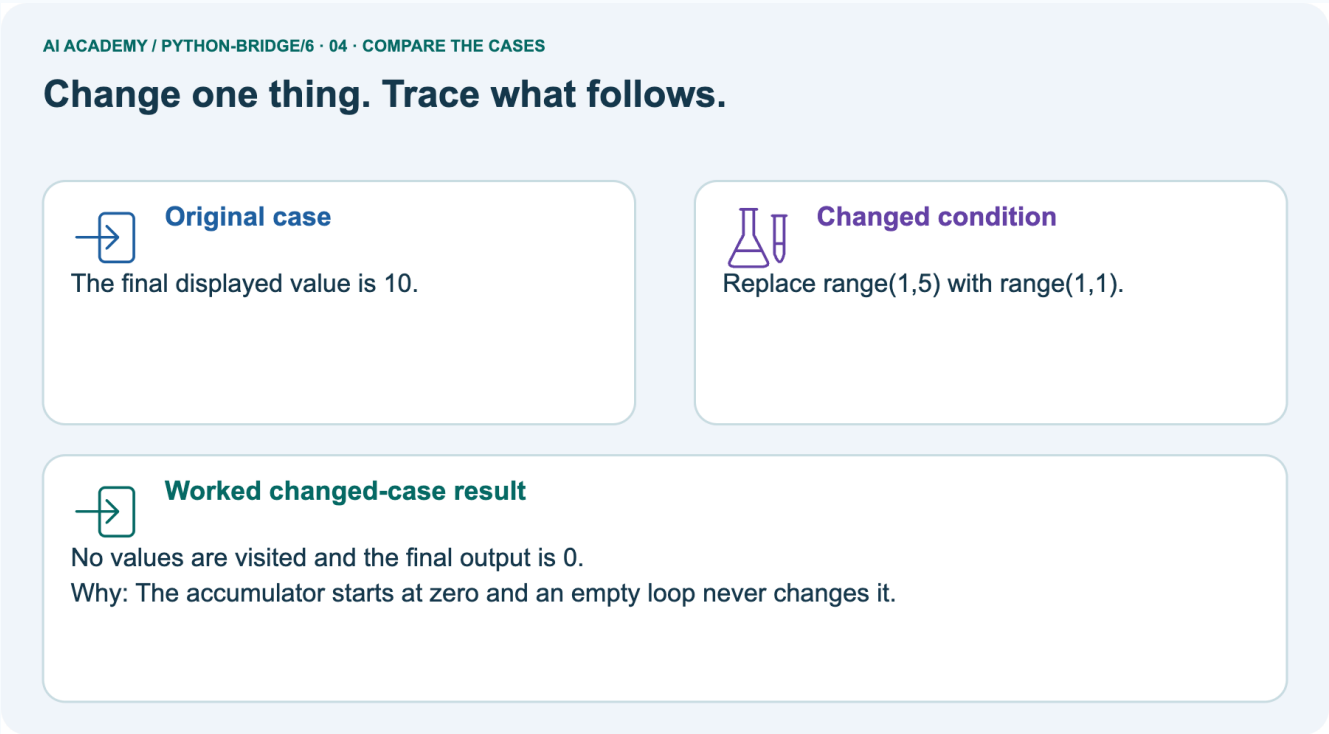
Running total after each visited value. Visit 1: Current value = 1, Running total = 1. Visit 2: Current value = 2, Running total = 3. Visit 3: Current value = 3, Running total = 6.

Visit 4: Current value = 4, Running total = 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

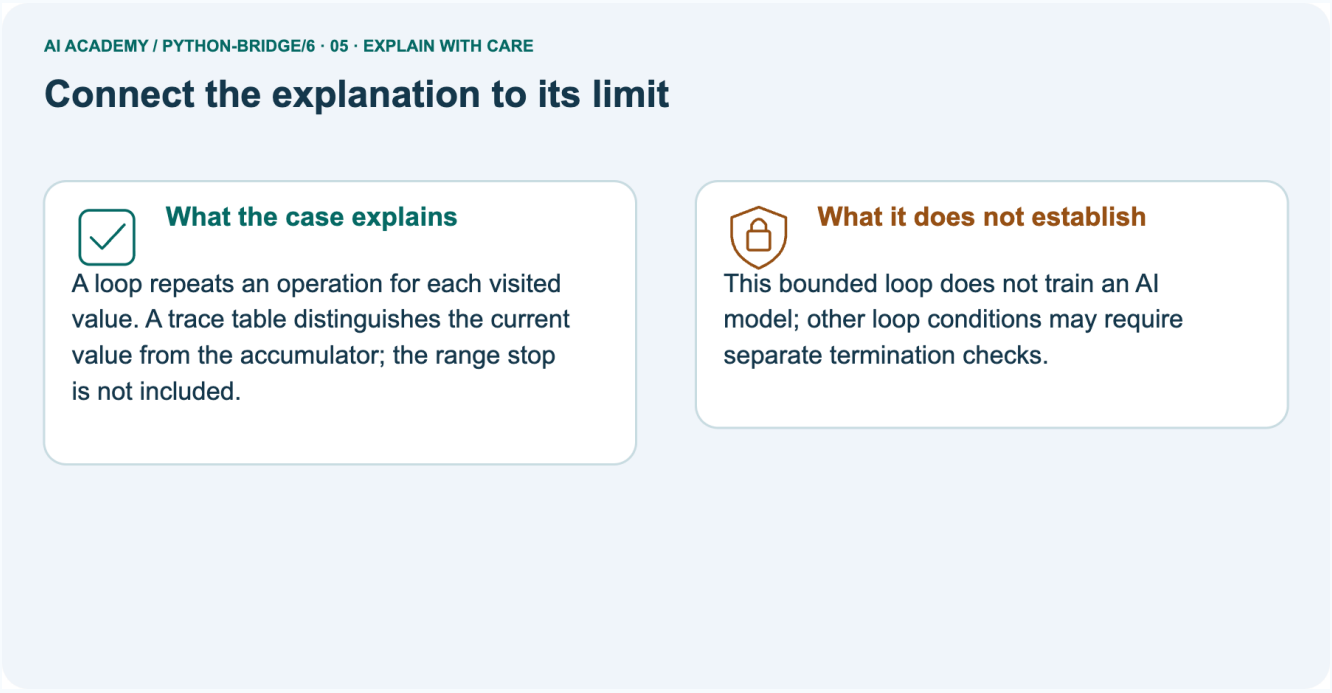
Original case: The final displayed value is 10. Changed condition: Replace range(1,5) with range(1,1). Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

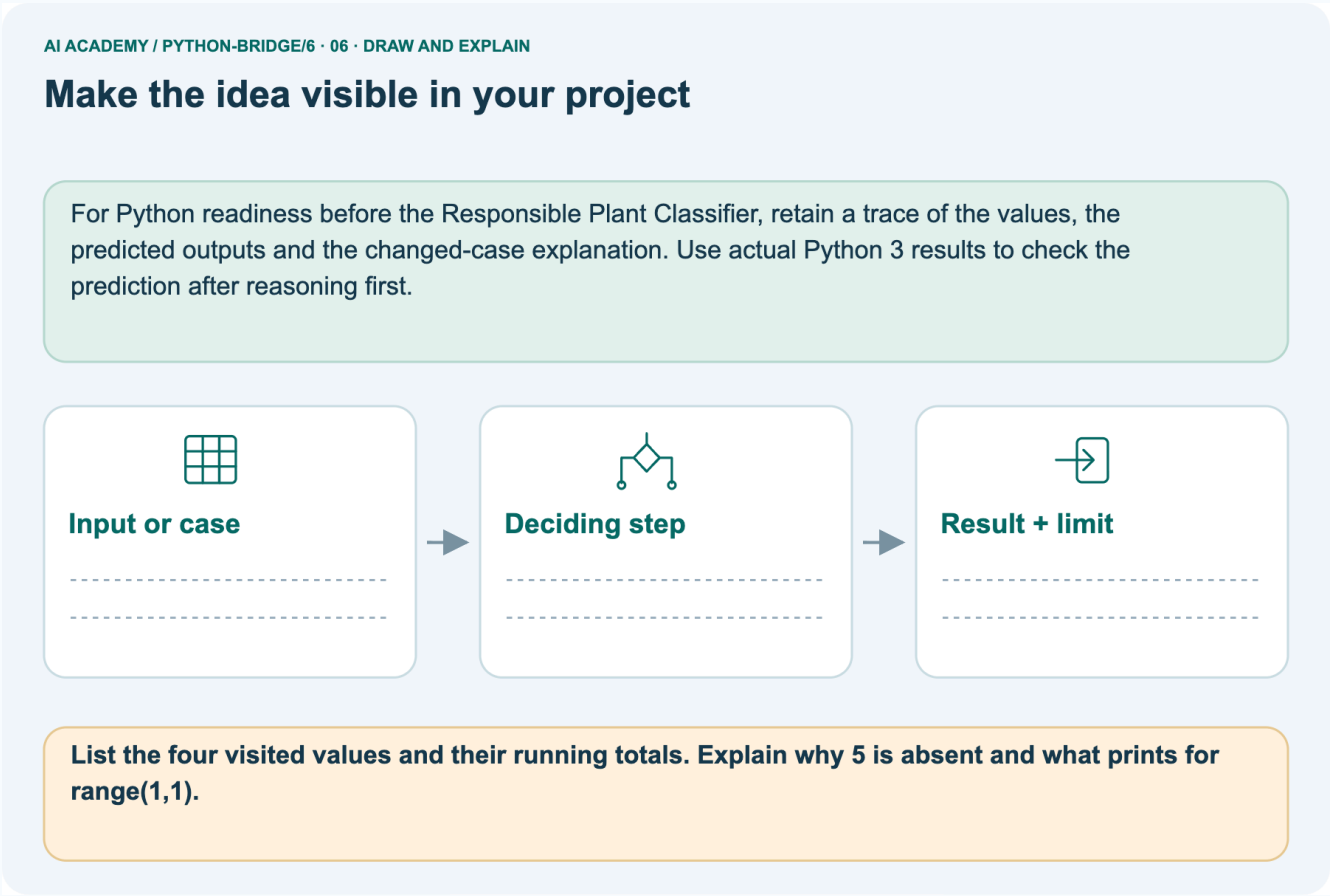
A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Limit: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Loops and trace tables: the case

02. Follow the deciding steps

03. Running total after each visited value

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.